Cover letter

Dear Editors,

Please consider the manuscript entitled **“VEHICLE-ODI: A Relational Decision Architecture for Orbital Debris Prioritization”** for publication in *Frontiers in Space Technologies*, Space Debris section, as a **Hypothesis and Theory** article.

The manuscript introduces VEHICLE-ODI (Orbital Debris Intelligence), a relational decision architecture for prioritizing orbital debris before observation, avoidance, stabilization, relocation or active debris removal resources are allocated. The central contribution is not a replacement for space situational awareness, astrodynamics, space traffic management or active debris removal. Rather, VEHICLE-ODI is proposed as an explainable decision-support layer that organizes debris objects, fragments, clouds, vulnerable assets and orbital regions as a structured graph under tension.

The paper contributes three elements that may be useful to the space debris community. First, it formalizes the prioritization problem as a relational tension problem rather than only an object-counting or object-scoring problem. Second, it proposes A1-A6 operational regimes to distinguish immediate conjunction pressure, recovered objects, fragmentation pressure, filtered fragments, dangerous rigid masses and diffuse fragmentation clouds. Third, it introduces an experimental Orbital Tension Index (OTI) for comparing structural stress across orbital regions under declared assumptions.

The manuscript is intentionally framed as a testable architecture. It proposes a synthetic simulation phase before operational use and clearly states its limitations: VEHICLE-ODI does not claim autonomous decision authority, does not replace existing orbital models, and does not constitute legal permission to intervene. Its purpose is to support transparent prioritization and structured discussion before expensive or irreversible action is considered.

This manuscript is original, is not under consideration elsewhere, and has not been published previously. No external funding was received. The author declares no conflict of interest.

Sincerely,

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